

Unit 33: Exponential Change and Separable Differential Equations — Full Solutions

Warm-up

1. $\frac{dy}{dx} = x$

$$dy = x \, dx \quad \Rightarrow \quad \int dy = \int x \, dx \quad \Rightarrow \quad y = \frac{x^2}{2} + C$$

2. $\frac{dy}{dx} = 2y$

Separate: $\frac{dy}{y} = 2 \, dx$.

$$\int \frac{dy}{y} = \int 2 \, dx \quad \Rightarrow \quad \ln |y| = 2x + C \quad \Rightarrow \quad y = Ae^{2x}$$

(where $A = e^C$ is just a renamed arbitrary constant).

3. $\frac{dy}{dx} = \frac{x}{y}$

Separate: $y \, dy = x \, dx$.

$$\int y \, dy = \int x \, dx \quad \Rightarrow \quad \frac{y^2}{2} = \frac{x^2}{2} + C \quad \Rightarrow \quad y^2 - x^2 = K$$

(renaming $2C$ as K).

4. $\frac{dy}{dx} = 3x^2$, $y(0) = 5$.

$$y = \int 3x^2 \, dx = x^3 + C$$

Apply $y(0) = 5$: $0 + C = 5 \Rightarrow C = 5$.

$$y = x^3 + 5.$$

5. $\frac{dy}{dx} = \frac{y}{x}$ ($x > 0$)

Separate: $\frac{dy}{y} = \frac{dx}{x}$.

$$\ln |y| = \ln |x| + C \quad \Rightarrow \quad y = e^{\ln |x| + C} = A \cdot x$$

(since $x > 0$, $|x| = x$).

6. $\frac{dy}{dx} = -2y$, $y(0) = 10$.

Separate: $\frac{dy}{y} = -2dx$.

$$\ln|y| = -2x + C \quad \Rightarrow \quad y = Ae^{-2x}$$

Apply $y(0) = 10$: $A = 10$.

$$y = 10e^{-2x}.$$

Standard

7. $\frac{dy}{dx} = xe^{-y}$

Separate: $e^y dy = x dx$.

$$\int e^y dy = \int x dx \quad \Rightarrow \quad e^y = \frac{x^2}{2} + C$$

Solve for y :

$$y = \ln\left(\frac{x^2}{2} + C\right)$$

8. $\frac{dy}{dx} = \frac{\cos x}{y}$

Separate: $y dy = \cos x dx$.

$$\int y dy = \int \cos x dx \quad \Rightarrow \quad \frac{y^2}{2} = \sin x + C \quad \Rightarrow \quad y^2 = 2\sin x + K$$

$$y = \pm\sqrt{2\sin x + K}$$

9. $\frac{dy}{dx} = y \cos x$

Separate: $\frac{dy}{y} = \cos x dx$.

$$\ln|y| = \sin x + C \quad \Rightarrow \quad y = Ae^{\sin x}$$

10. $\frac{dy}{dx} = x^2 y$, $y(0) = 2$.

Separate: $\frac{dy}{y} = x^2 dx$.

$$\ln|y| = \frac{x^3}{3} + C \quad \Rightarrow \quad y = Ae^{x^3/3}$$

Apply $y(0) = 2$: $A = 2$.

$$y = 2e^{x^3/3}.$$

11. $\frac{dy}{dx} = 2xy^2$

Separate: $\frac{dy}{y^2} = 2x dx$.

$$\int y^{-2} dy = \int 2x dx \Rightarrow -y^{-1} = x^2 + C$$

Solve for y :

$$y = \frac{-1}{x^2 + C}$$

12. $\frac{dy}{dx} = e^{x+y}$

Rewrite: $e^{x+y} = e^x \cdot e^y$. Separate: $\frac{dy}{e^y} = e^x dx \Rightarrow e^{-y} dy = e^x dx$.

$$\int e^{-y} dy = \int e^x dx \Rightarrow -e^{-y} = e^x + C$$

Solve for y : $e^{-y} = -e^x - C$. Renaming the constant so the right side reads more cleanly ($K = -C$):

$$e^{-y} = K - e^x \Rightarrow -y = \ln(K - e^x) \Rightarrow y = -\ln(K - e^x)$$

Challenge

13. $\frac{dy}{dx} = \sin x \cdot e^{-y}$

Separate: $e^y dy = \sin x dx$.

$$\int e^y dy = \int \sin x dx \Rightarrow e^y = -\cos x + C$$

Solve for y (renaming the constant for a cleaner form):

$$y = \ln(C - \cos x)$$

14. Deriving the Law of Exponential Change from $\frac{dy}{dt} = ky$.

Separate: $\frac{dy}{y} = k dt$.

$$\int \frac{dy}{y} = \int k dt \Rightarrow \ln|y| = kt + C$$

Exponentiate both sides:

$$y = e^{kt+C} = e^C \cdot e^{kt}$$

Let $C_1 = e^C$ (a renamed positive constant):

$$y = C_1 e^{kt}$$

Relating C_1 to y_0 : plug in $t = 0$:

$$y(0) = C_1 e^{k(0)} = C_1 \cdot 1 = C_1$$

So C_1 is exactly $y(0)$, the initial value — which we usually call y_0 . This confirms:

$$y = y_0 e^{kt}$$

15. $\frac{dy}{dx} = y^2 \sin x, y(0) = 1.$

Separate: $\frac{dy}{y^2} = \sin x dx.$

$$\int y^{-2} dy = \int \sin x dx \Rightarrow -y^{-1} = -\cos x + C$$

Multiply both sides by -1 : $y^{-1} = \cos x - C$. Rename the constant ($-C \rightarrow C$ again, standard practice):

$$\frac{1}{y} = \cos x - C \Rightarrow y = \frac{1}{\cos x - C}$$

Apply $y(0) = 1$: $1 = \frac{1}{\cos(0) - C} = \frac{1}{1 - C}$, so $1 - C = 1 \Rightarrow C = 0$.

$$y = \frac{1}{\cos x} = \sec x$$

$$y = \sec x.$$

16. $\frac{dy}{dx} = \frac{x+1}{y-2}, y(0) = 4.$

Separate: $(y-2) dy = (x+1) dx.$

$$\int (y-2) dy = \int (x+1) dx \Rightarrow \frac{y^2}{2} - 2y = \frac{x^2}{2} + x + C$$

Apply $y(0) = 4$: $\frac{16}{2} - 2(4) = 0 + 0 + C \Rightarrow 8 - 8 = C \Rightarrow C = 0$.

$$\frac{y^2}{2} - 2y = \frac{x^2}{2} + x$$

Multiply by 2: $y^2 - 4y = x^2 + 2x$. Solve for y using the quadratic formula (treating this as $y^2 - 4y - (x^2 + 2x) = 0$):

$$y = \frac{4 \pm \sqrt{16 + 4(x^2 + 2x)}}{2} = 2 \pm \sqrt{4 + x^2 + 2x} = 2 \pm \sqrt{x^2 + 2x + 4}$$

Check which sign matches $y(0) = 4$: $2 \pm \sqrt{0 + 0 + 4} = 2 \pm 2$. We need 4, so we take the + sign.

$$y = 2 + \sqrt{x^2 + 2x + 4}.$$

17. $\frac{dy}{dx} = -xy$, $y(0) = 3$.

Separate: $\frac{dy}{y} = -x dx$.

$$\ln|y| = -\frac{x^2}{2} + C \Rightarrow y = Ae^{-x^2/2}$$

Apply $y(0) = 3$: $A = 3$.

$$y = 3e^{-x^2/2}.$$

Applied

18. $\frac{dP}{dt} = 0.04P$, $P(0) = 1000$.

By the Law of Exponential Change:

$$P(t) = 1000e^{0.04t}$$

At $t = 20$:

$$P(20) = 1000e^{0.8} \approx 1000(2.2255) \approx 2225.5$$

Answer: $P(t) = 1000e^{0.04t}$, and after 20 years, the population is ≈ 2226 .

19. $\frac{dT}{dt} = -0.1(T - 70)$, $T(0) = 200$.

Using the Newton's Law of Cooling shortcut, with $T_s = 70$ and $T_0 - T_s = 200 - 70 = 130$:

$$T(t) = 70 + 130e^{-0.1t}$$

At $t = 10$:

$$T(10) = 70 + 130e^{-1} \approx 70 + 130(0.3679) \approx 70 + 47.83 \approx 117.83$$

Answer: $T(t) = 70 + 130e^{-0.1t}$, and after 10 minutes, the temperature is $\approx 117.8^\circ$.

20. $\frac{dA}{dt} = -0.05A$, $A(0) = 200$.

$$A(t) = 200e^{-0.05t}$$

Find t when $A = 100$:

$$100 = 200e^{-0.05t} \Rightarrow 0.5 = e^{-0.05t} \Rightarrow \ln(0.5) = -0.05t$$

$$t = \frac{\ln(0.5)}{-0.05} = \frac{-0.6931}{-0.05} \approx 13.86$$

Answer: $A(t) = 200e^{-0.05t}$, and it takes ≈ 13.86 time units to decay to 100 grams.

21. $\frac{dC}{dt} = -0.3C$, $C(0) = 80$.

$$C(t) = 80e^{-0.3t}$$

At $t = 5$:

$$C(5) = 80e^{-1.5} \approx 80(0.22313) \approx 17.85$$

Answer: $C(t) = 80e^{-0.3t}$, and after 5 hours, the concentration is ≈ 17.85 mg.